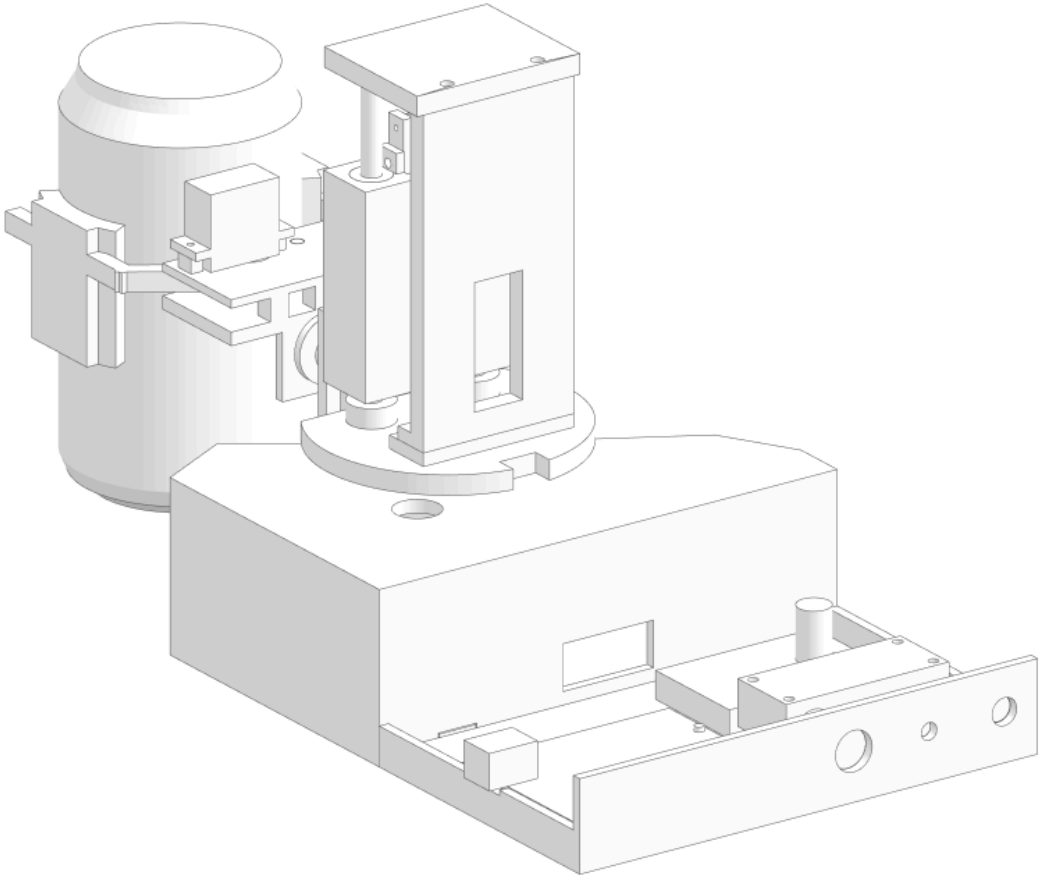
CONCEPT A

Can-flipping robot · assembly manual · MF 2331/I



1 person · ~3 h after printing · read each page, then build. **Servo angles:** load `firmware/servo_setup` and type the command in the round badge (Serial Monitor, 9600).

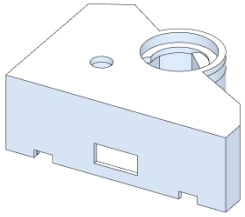
You need

Tool
Phillips screwdrivers #1 and #2
Multimeter
Hot-glue gun
Superglue
Wire stripper
Small flat screwdriver (buck pot)
Hacksaw (6 mm rod) + file
Vise or clamp (bearing press)
12 mm drill bit, turned by hand
Sandpaper 120 + 240, craft knife, flush cutters

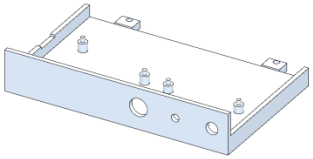
Hardware	Qty
 #6-32 × ½″	8
 #8-32 × ½″	4
 servo-bag screws	~14
6807-2RS bearing	1
LM6UU linear bearing	4
6 mm rod, cut 2 × 112 mm	2
Servos: MG996R / MG90S / SG90	1 / 2 / 1

! Before building: bench-test the electronics (docs/13_electronics_bench_test.md) and set the buck to **5.8 V**.

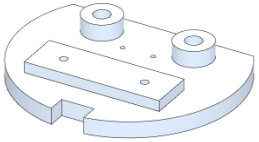
Printed parts



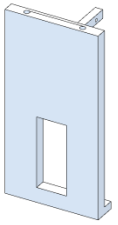
CA-101
Base



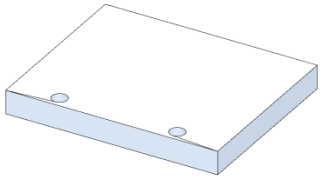
CA-102
Tray



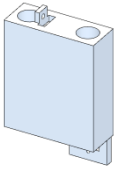
CA-201
Turret plate



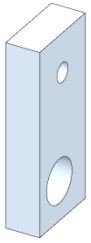
CA-202
Spine



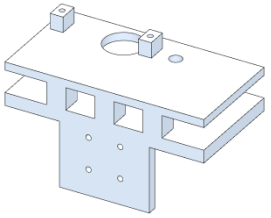
CA-203
Top cap



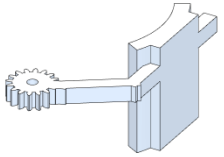
CA-301
Carriage arm



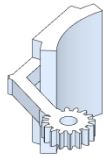
CA-302
Lift link



CA-401
Gripper frame



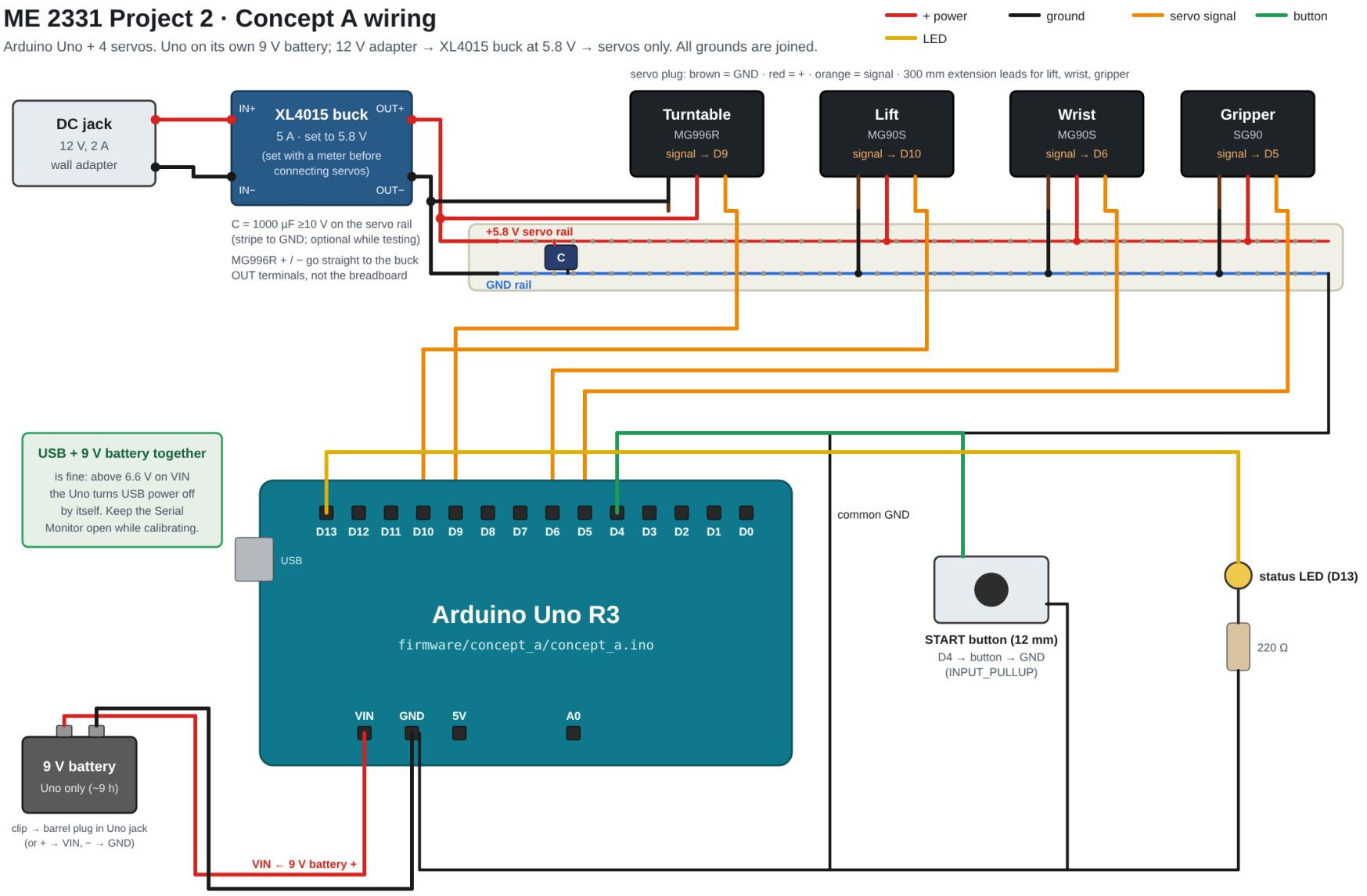
CA-402
Finger R



CA-403
Finger L

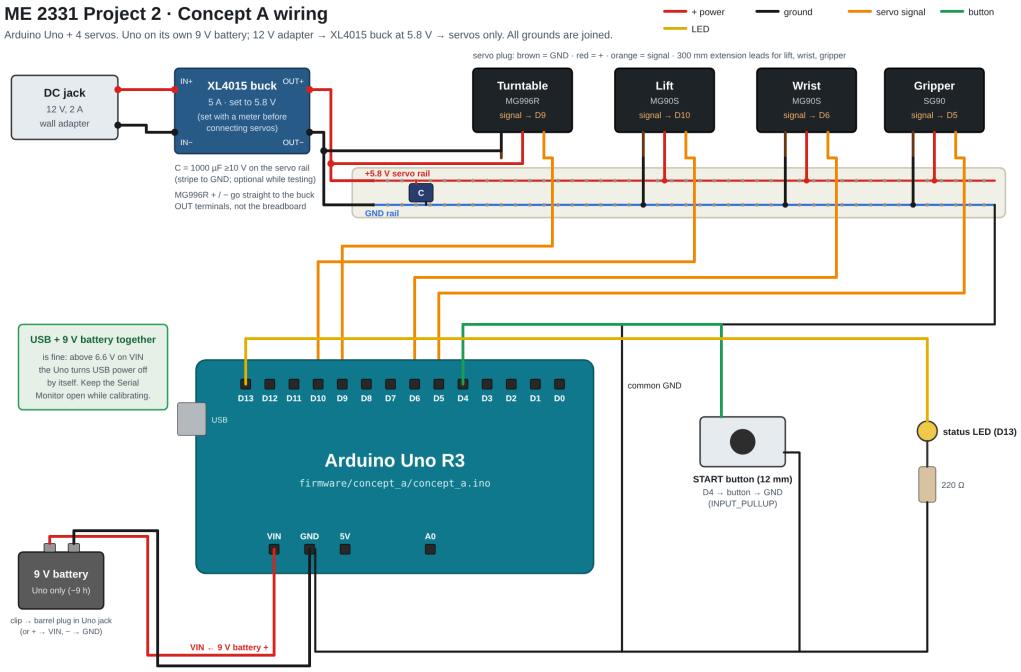
PART E

Electronics first: wire it flat on the table and test it while the parts print



E1

Circuit: every connection



	From	To	Wire	Notes
☐	9 V battery +	Uno barrel jack + (or **VIN**)	red	Uno power only. Snap clip with a 5.5 × 2.1 mm plug
☐	9 V battery −	Uno barrel jack − (or **GND**)	black	
☐	12 V jack +	XL4015 IN+	red	Servo power only. "Jack" = the DC panel jack or the LIZVOR terminal-block tip
☐	12 V jack −	XL4015 IN−	black	
☐	XL4015 OUT+ / OUT−	servo rail + / GND, and the MG996R's red / brown	red / black	5.8 V. MG996R power on the terminals, not the breadboard
☐	1000 µF capacitor	across the servo rail, stripe (−) to GND	—	Optional while testing; fit before the final runs
☐	GND rail	Uno GND	black	**common ground: required** (the servo signals need it)
☐	Turntable servo MG996R	signal → **D9** (+ / − at the buck terminals)	orange	One extension lead is cut to split its power from its signal
☐	Lift servo MG90S	signal → **D10**	orange	300 mm extension lead
☐	Wrist servo MG90S	signal → **D6**	orange	300 mm lead. Code widens its pulse range to 500–2500 µs to reach 180°
☐	Gripper servo SG90	signal → **D5**	orange	500 mm extension lead (or a taped jumper chain, see 11_bom)
☐	START button (12 mm)	**D4** → button → GND	green / black	`INPUT_PULLUP`, debounced in code
☐	Status LED	**D13** → 220 Ω → LED → GND	yellow / black	off = idle, blinking = moving, on = done

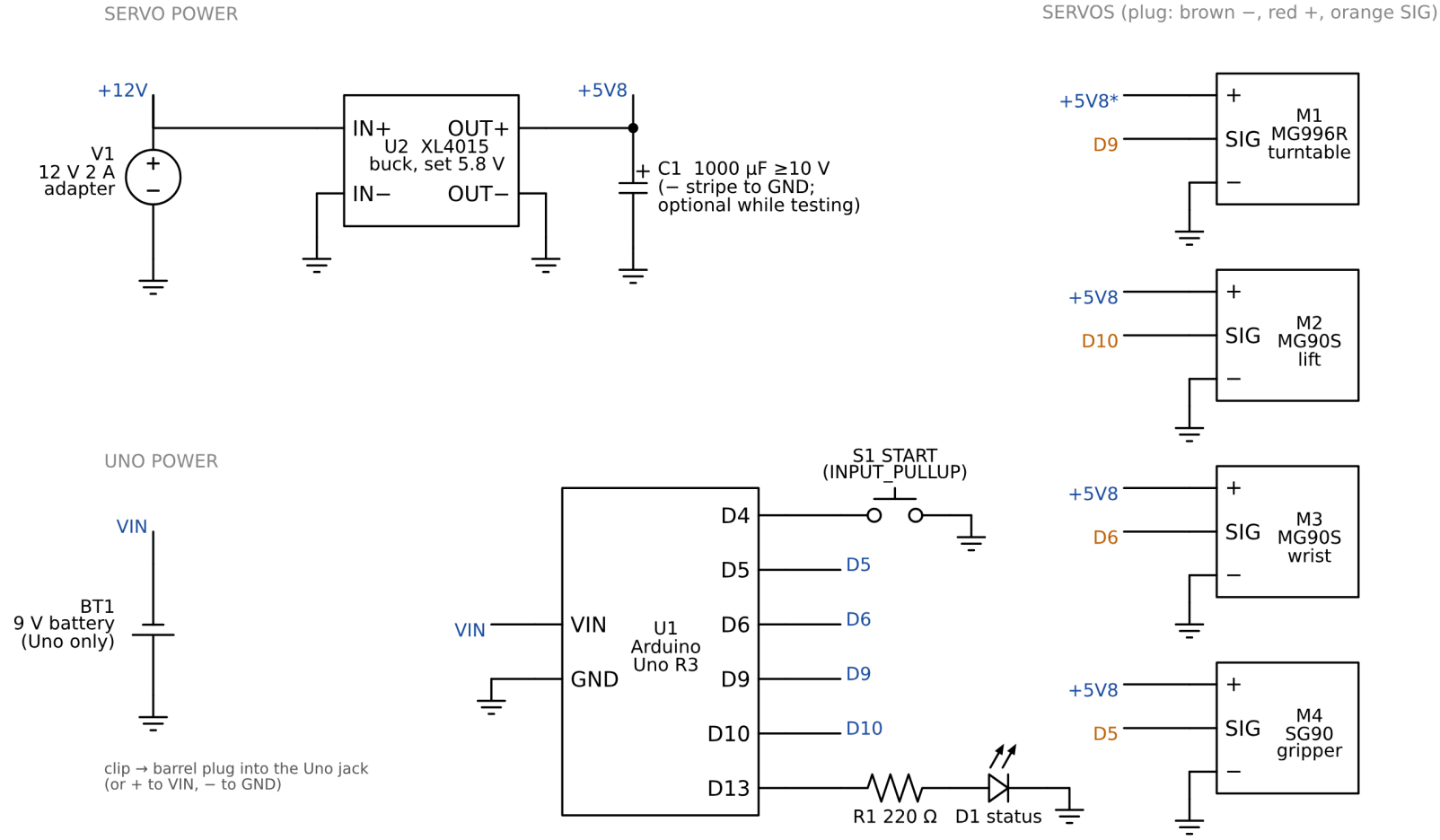
Never connect the 5.8 V rail to the Uno's 5V pin: the Uno gets its power only from the 9 V battery. Servo plug: brown = GND, red = +, orange = signal.

E1b

Schematic (standard symbols): same circuit, nets with the same name are connected

ME 2331 Project 2 · Concept A schematic

Net labels with the same name are connected; every ground symbol is the same GND (servo GND and Uno GND joined).



* M1 + / − go straight to the buck OUT terminals, not through the breadboard (2.5 A stall).

Uno on its own 9 V battery; the 12 V adapter feeds only the buck (servos). Uno GND and servo GND must be joined.

E2

Bench test 1: power, no servos yet

	#	Do	Check	Expected
☐	1	LIZVOR adapter, nothing connected, plain 5.5 × 2.1 mm tip	Display + meter on the tip (centre = +)	**12.0 V**, centre positive. Tape the knob
☐	2	Adapter → terminal-block tip → buck **IN+ / IN−**	Buck display / meter on **OUT+ / OUT−**	Turn the buck pot (many turns) to **5.8 V**. Dot of nail polish on the pot
☐	3	Buck OUT → breadboard **+ / −** rails. 1000 µF capacitor across the rails	Stripe on the capacitor = **−** rail	Rail reads 5.8 V
☐	4	Unplug. Uno on **USB only**, upload `bench_test`	Uno LED	Blinks 3×
☐	5	**9 V battery** → Uno barrel jack (snap clip with a 5.5 × 2.1 mm plug). Meter on the battery first	Unplug USB, plug the battery	Battery ≥ 8.5 V (fresh ~9.3 V); LED blinks 3× again (running on the battery)
☐	5b	Uno **GND** → rail **−** (common ground: required)	Meter between Uno GND and the buck OUT−	0 Ω / beeps
☐	6	START button: D4 ↔ GND. LED: D13 → 220 Ω → LED → GND	Press START	LED lights while pressed

Never connect the rail (5.8 V) to the Uno's 5V pin. The Uno gets its power only from the 9 V battery (jack or VIN); the 12 V adapter feeds only the buck. The 1000 µF capacitor is optional while testing: skip row 3 if you don't have one yet, and fit it before the final runs.

E3

Bench test 2: one servo at a time (firmware/bench_test)

Plug only the servo for the next test. Brown = GND (rail −), red = + (rail +), orange = signal (pin). The MG996R's + / − go to the **buck OUT terminals** (cut extension), its signal to D9.

	Test	Servo	Plug signal to	Press START, expect
☐	1	SG90 gripper	D5	Goes to 20°, wiggles ±10°, back
☐	2	MG90S wrist	D6	0°, wiggle to 20°, then **a full half-turn 0 → 180 → 0**: mark the horn and check it really reaches 180°
☐	3	MG90S lift	D10	30°, wiggles ±10°
☐	4	MG996R turntable	D9 (+/− on the buck)	135°, wiggles ±10°
☐	5	All four plugged in	—	All at HOME for 5 s. **Meter on the rail: stays above 5.5 V**

Each test: smooth, no buzzing, the Uno doesn't reset (no 3× blink in the middle).

E3b

Laptop control panel (viewer/control_panel.html)

1. Upload `firmware/concept_a/concept_a.ino` with the Arduino IDE, then **close the IDE's Serial Monitor** (only one program can use the USB port). 2. Open `viewer/control_panel.html` in **Chrome or Edge** (or the project site link). Click **Connect to Arduino** and pick the Uno's port. The Uno restarts; after ~2 s the angles appear. 3. Each servo has a slider, ±5° buttons and its working angles (base START 135° / TARGET 45°, lift DOWN / UP, wrist UPRIGHT / FLIPPED, grip OPEN / RELEASE / CLOSED). Every move is smooth and stays inside safe limits. 4. **HOME** brings the robot home one servo at a time. **START** runs the whole sequence (same as the physical button, which keeps working). START only runs from HOME. **STOP** (or Esc) holds where it is.

	Check	**Expected**
☐	Each servo button	Only that servo moves, to that angle, smoothly
☐	HOME	Lift up → wrist → base → grip → lift down
☐	START at HOME	GRIP → LIFT → FLIP → TURN → LOWER → RELEASE, ends DONE
☐	START again after DONE (can removed)	Goes home
☐	STOP during a move	Holds; HOME before the next START

Tested without hardware: the firmware's commands on a simulated Uno (`backend/tools/firmware_test/run.sh`, 15 checks) and the page driving the live firmware through a fake USB port (`panel_test.js`, 11 checks).

E4

If something is wrong

	Symptom	Likely cause	Fix
☐	Uno resets (blinks 3×) when a servo moves	9 V battery flat, or the servo supply wired to the Uno by mistake	Battery < 7.5 V: replace it. Check nothing from the rail goes to the Uno's 5V or VIN
☐	Servos twitch or don't respond at all	No common ground	Uno GND ↔ rail − (row 5b)
☐	Servo buzzes / jitters	Signal without ground, loose jumper	Re-seat; shorter jumpers; ground first
☐	Servo doesn't move at all	Signal on the wrong pin, dead servo	Swap the plug to a known-good test; try the spare
☐	One servo works sometimes, then stops responding	Cable / dupont joint (most often), power at its plug, or the servo	**Swap test** (unmounted, horn off): plug that servo into a good servo's cable, and a good servo into its cable. Problem follows the servo → servo; stays with the cable / pin → wiring; both fine → it was a loose joint. Same code drives all four, so code alone is unlikely
☐	Wrist stops short of 180°	That MG90S's range	`servo_setup`: try `WRIST_MIN_US` / `WRIST_MAX_US` 450 / 2550; else swap MG90S (buy a spare)
☐	Rail below 5.5 V in test 5	Weak adapter / long thin wires / no capacitor	Shorter rail wires; check the adapter is at 12.0 V under load; fit the 1000 µF
☐	Buck gets hot	Pot set too high, short	Unplug, re-check OUT = 5.8 V with nothing connected

E5

Wiring dry run (decides what to return)

With the servos still loose, build the **jumper chains** (prototype v1 BOM): 2 jumpers per wire for the gripper, 1 for the wrist and lift, tape every joint. Re-run tests 1–5 through the chains.

• All pass, joints stay put when you tug and bend them → **return the extension leads**.

• A joint drops out or a servo resets → keep the extension leads.

Record the results (angles that looked right, rail voltage in test 5) in the build log: they go in the report's test section.

Then: PART P, printing, and PART M, the mechanics

Print the economy plates (cad/print/plates_eco, next pages), fix up the fits, then follow steps 1–23. Keep the bench-test wiring: it moves into the tray in step 21.

P1

Print settings: only the quality each part needs

	Profile	Layer	Walls	Top / bottom	Infill	Speeds (outer / inner / infill, mm/s)	For
☐	**DRAFT**	0.28 mm	2	3 / 3	10% grid	45 / 60 / 90	big, hidden, lightly loaded: coarse layers, 2–3 walls, 10–15 % infill, faster speeds
☐	**STANDARD**	0.2 mm	3	4 / 4	20% gyroid	35 / 50 / 70	has press fits or carries a load: 0.2 mm, 3 walls, 20 % infill
☐	**DETAIL**	0.2 mm	4	5 / 5	100% rectilinear	25 / 35 / 50	gear teeth: slow outer walls for an accurate tooth shape, solid

	Part	Profile	Changes from the profile	Before	After	Must be accurate	Can be rough
☐	CA-101 Base	**DRAFT**	perimeters 3, supports (from the bed)	12 h 47, 139 g	**6 h 33, 124 g**	6807 bearing pocket (47.1 mm), MG996R screw holes	everything else: it is a box. Layer lines, rough sides, small sag in the pockets
☐	CA-102 Electronics tray	**DRAFT**	—	7 h 08, 66 g	**3 h 01, 50 g**	nothing critical: board pegs and 2 screw tabs	everything
☐	CA-201 Turret plate	**STANDARD**	supports (from the bed)	3 h 08, 31 g	**2 h 23, 28 g**	hub into the bearing bore (34.95 mm), 2 rod holes (6.0 mm), horn screw holes	the plate's top face and edges
☐	CA-202 Spine	**DRAFT**	perimeters 3, fill density 15%	2 h 25, 22 g	**1 h 05, 18 g**	lift-servo screw holes, 2 screw holes at each end	faces and edges
☐	CA-203 Top cap	**DRAFT**	fill density 15%	1 h 09, 10 g	**29 min, 8 g**	2 blind rod holes (6.0 mm)	everything else
☐	CA-301 Carriage + arm	**STANDARD**	supports (from the bed)	4 h 14, 47 g	**3 h 08, 42 g**	4 LM6UU pockets (12.15 mm) in line, wrist-servo screws, link pivot hole	arm faces
☐	CA-302 Lift link	**STANDARD**	fill density 100%, fill pattern rectilinear, brim width 5	5 min, 1 g	**5 min, 1 g**	the 2 holes	-
☐	CA-401 Gripper frame	**STANDARD**	fill density 30%, external perimeter speed 25, perimeters 4	1 h 39, 17 g	**1 h 47, 19 g**	gear axle hole and servo position: they set the gear mesh	flange faces
☐	CA-402 Finger, right	**DETAIL**	supports (from the bed)	1 h 41, 23 g	**1 h 55, 23 g**	gear teeth, axle bore 3.9 mm	pad back face (it gets the silicone sheet)
☐	CA-403 Finger, left	**DETAIL**	supports (from the bed)	1 h 42, 23 g	**1 h 57, 24 g**	gear teeth, servo-spline socket	pad back face
☐	CA-900 Tolerance coupon	**STANDARD**	—	1 h 02, 8 g	**48 min, 8 g**	it is the test of the fits: print it with the STANDARD profile	-
☐	**Total**			**36 h 59, 387 g**	**23 h 12, 345 g**		

Import `concept_a_PETG_prusaslicer.ini`, open a `*_eco.3mf` plate (next page): each part's profile is already inside. Coupon first.

P2

Plates, then fix the fits (sandpaper, drill, knife)

	Plate	Parts	Time	PETG
☐	`plate0_coupon_eco.3mf`	CA-900	48 min	8 g
☐	`plate1_base_eco.3mf`	CA-101 CA-201 CA-203 CA-302	9 h 30	161 g
☐	`plate2_tray_spine_eco.3mf`	CA-102 CA-202 CA-402	6 h 03	92 g
☐	`plate3_arm_claw_eco.3mf`	CA-301 CA-401 CA-403	6 h 55	84 g
☐	`planP_1_claw_eco.3mf`	CA-401 CA-402	3 h 43	42 g
☐	`planP_2_finger_eco.3mf`	CA-403 CA-900	2 h 45	31 g

Tools: sandpaper 120 + 240 grit, a 6 mm and a 5/32 in (4 mm) drill bit turned by hand, a 12 mm dowel or marker wrapped in sandpaper, a craft knife, flush cutters / pliers for supports.

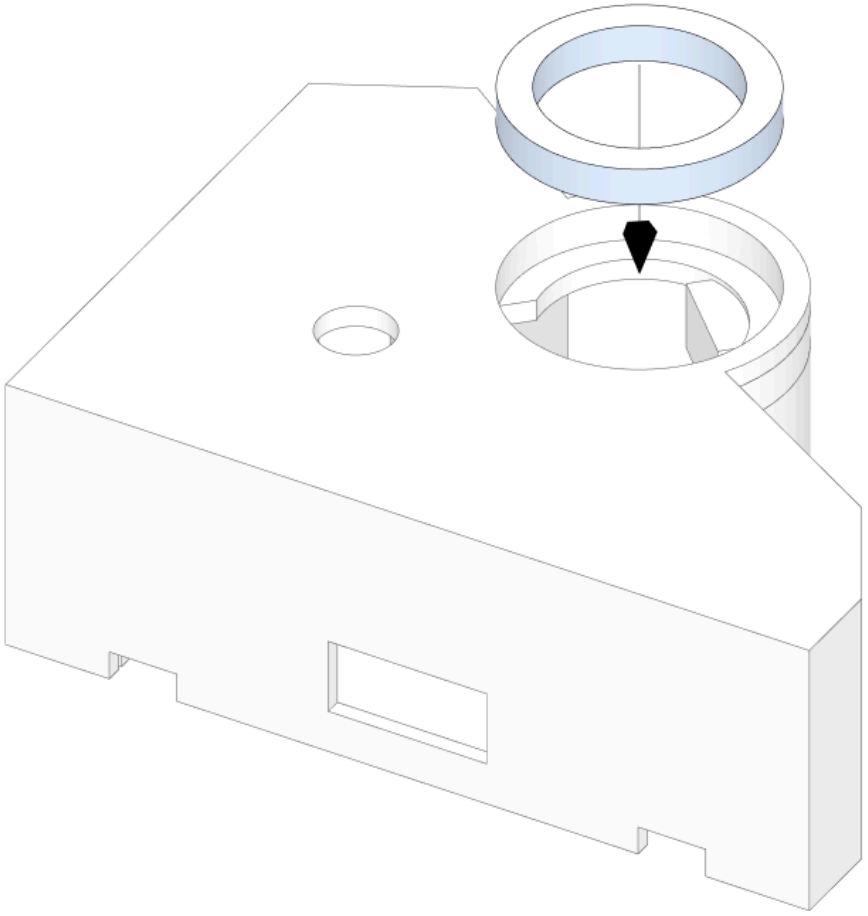
	Part	Fix
☐	CA-101 Base	Bearing tight: sand the pocket wall (120 grit on a finger) until it presses in with thumbs. Loose: one layer of tape. Clear support bits out of the servo pocket with pliers
☐	CA-102 Electronics tray	Pegs too fat for the Uno / buck holes: twist the board on, or shave a peg with a knife. Snapped peg: a dab of hot glue holds the board
☐	CA-201 Turret plate	Hub tight in the bearing: sand the hub evenly, turning it. Rod holes tight: drill 6 mm (or a 15/64 in bit) by hand
☐	CA-202 Spine	Servo screws: run each screw in once and back out before final assembly
☐	CA-203 Top cap	Rod won't go in: drill 6 mm, 8 mm deep (tape flag on the bit)
☐	CA-301 Carriage + arm	Pockets print on their side, so their top droops: ream with 12 mm drill or sandpaper on a 12 mm dowel / marker until the LM6UU presses in; check the carriage slides down both rods by its own weight
☐	CA-302 Lift link	Tiny part: peel the brim, open the pivot hole with a 5/32 in (4 mm) drill if the screw binds
☐	CA-401 Gripper frame	Don't open the axle hole more than needed: the gears must mesh without play
☐	CA-402 Finger, right	Remove supports, then file only the pad area; never the teeth. Bore tight: twist a 5/32 in drill through by hand
☐	CA-403 Finger, left	Spline socket tight: scrape lightly inside with the knife tip, test the SG90 spline after every pass. Never file the teeth
☐	CA-900 Tolerance coupon	Coarse (draft) layers can make holes come out slightly tight: expect a bit more sanding / drilling on draft parts

Rules: sand a press fit a little at a time and test after every few strokes (you can't add plastic back; if it gets loose, one layer of tape or a drop of CA glue). Never sand or file gear teeth. A draft part that cracks at a screw: the screw was forced: drill the hole one size bigger and use the next screw size or hot glue.

1

Bearing into the base

PART M

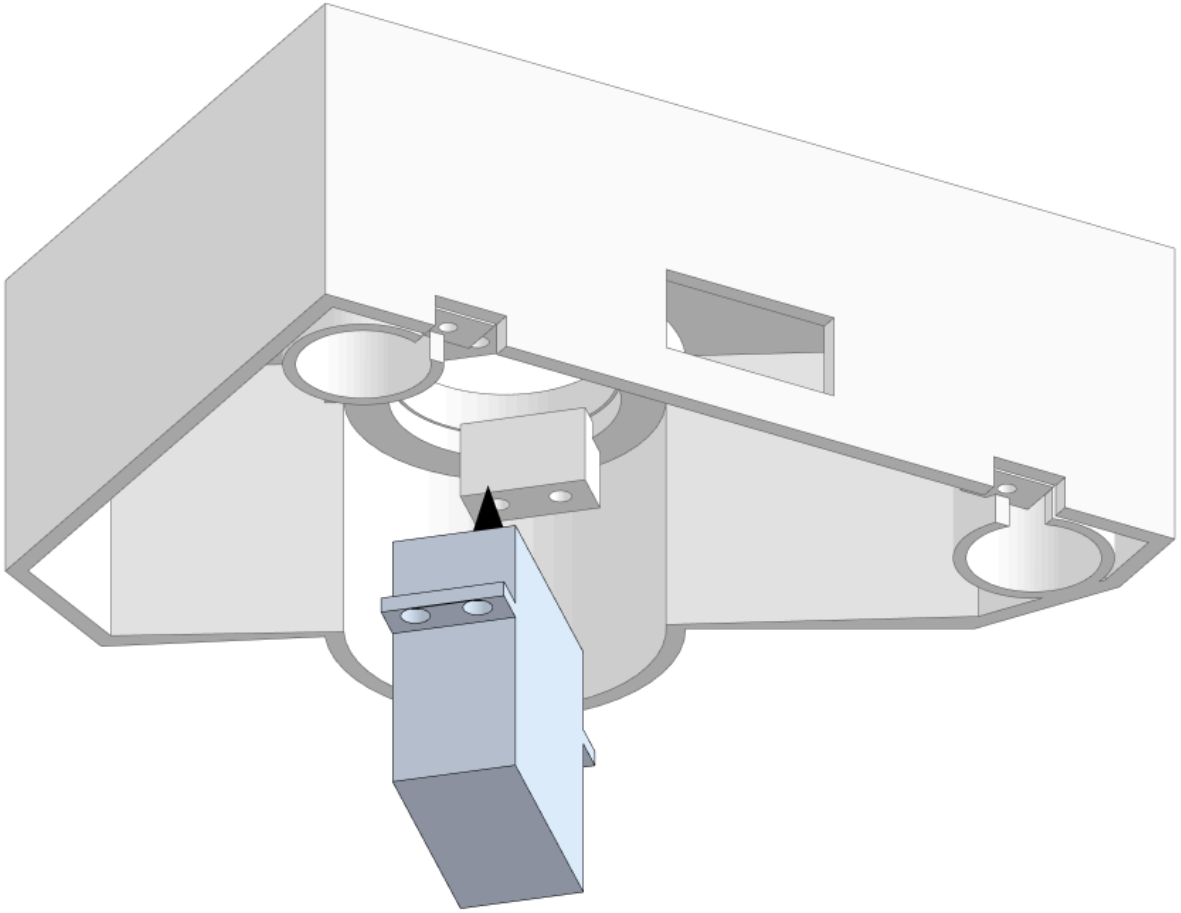


Press on the outer ring only, evenly (flat block + vise)

2

Turntable servo

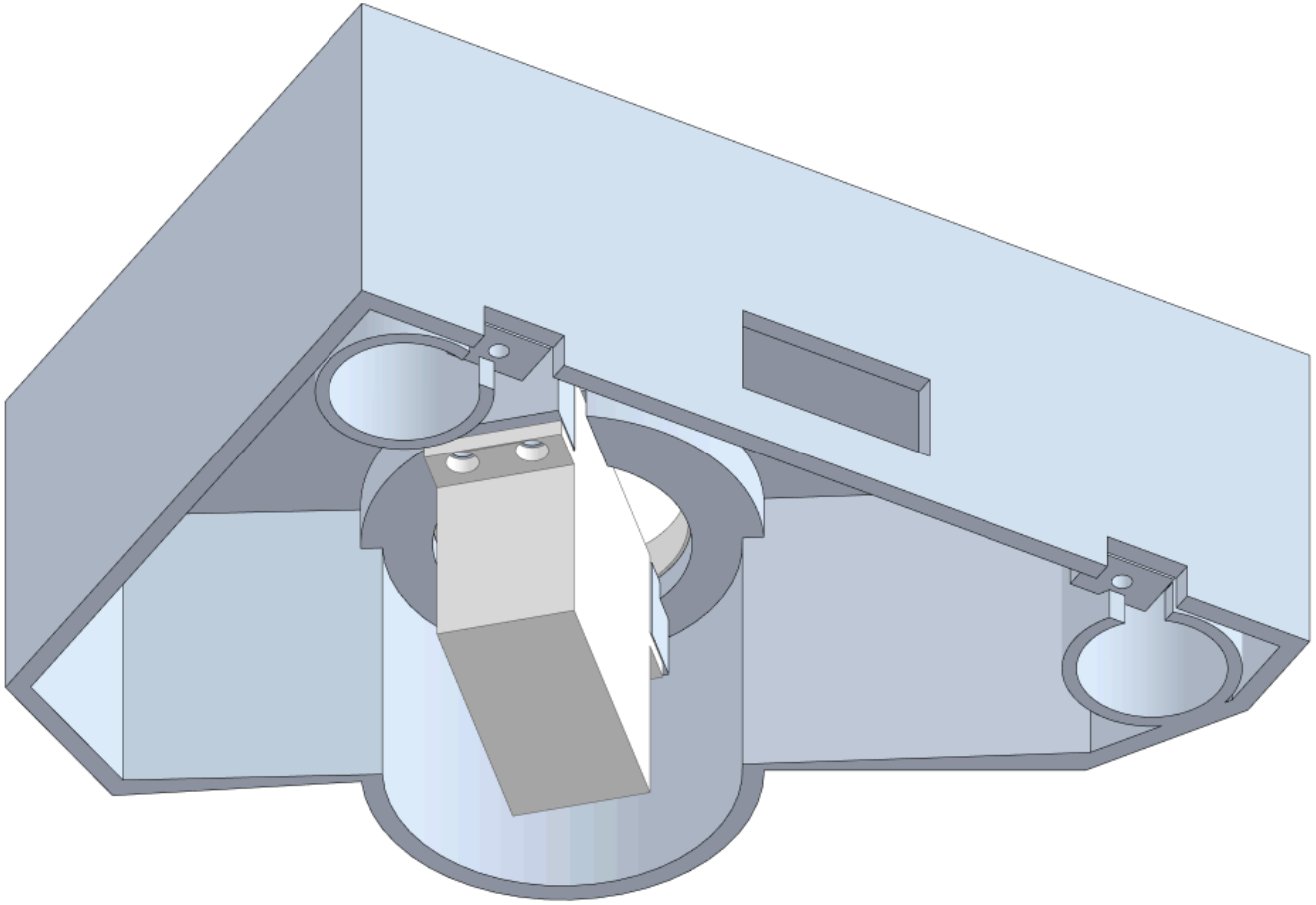
4× #8-32 × ½″



From underneath; body points away from the arm side

3

Weight + feet

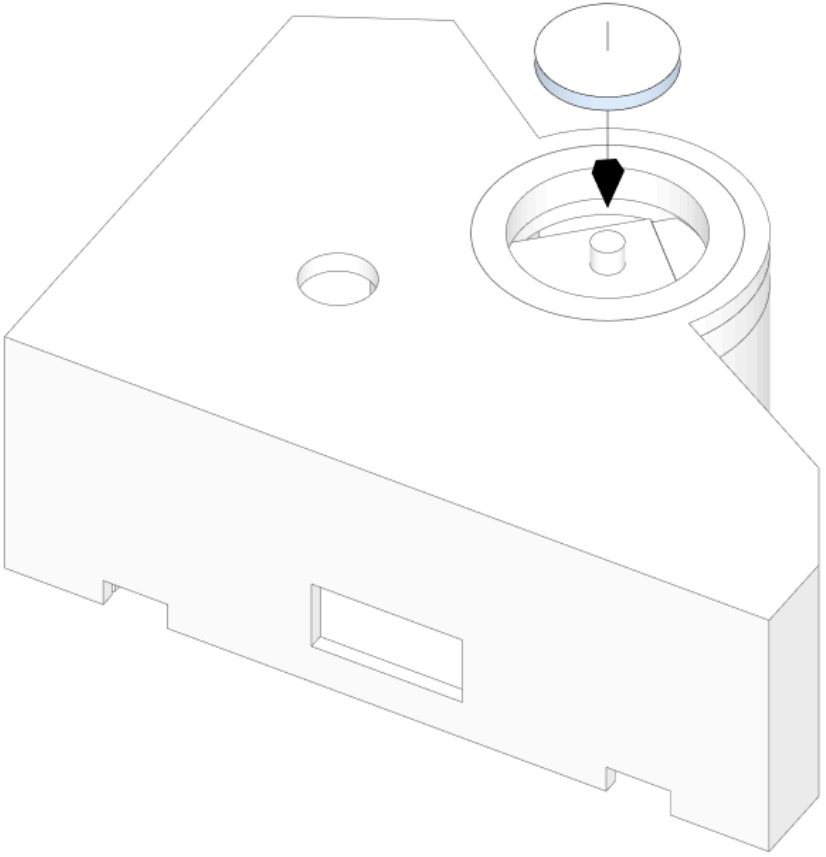


(!) ~100 g in the 2 back pockets (washers, pennies, rocks): GLUE or TAPE it in. 4 feet or hot-glue dots

4

Servo horn

1× servo-bag



MG996R

135°

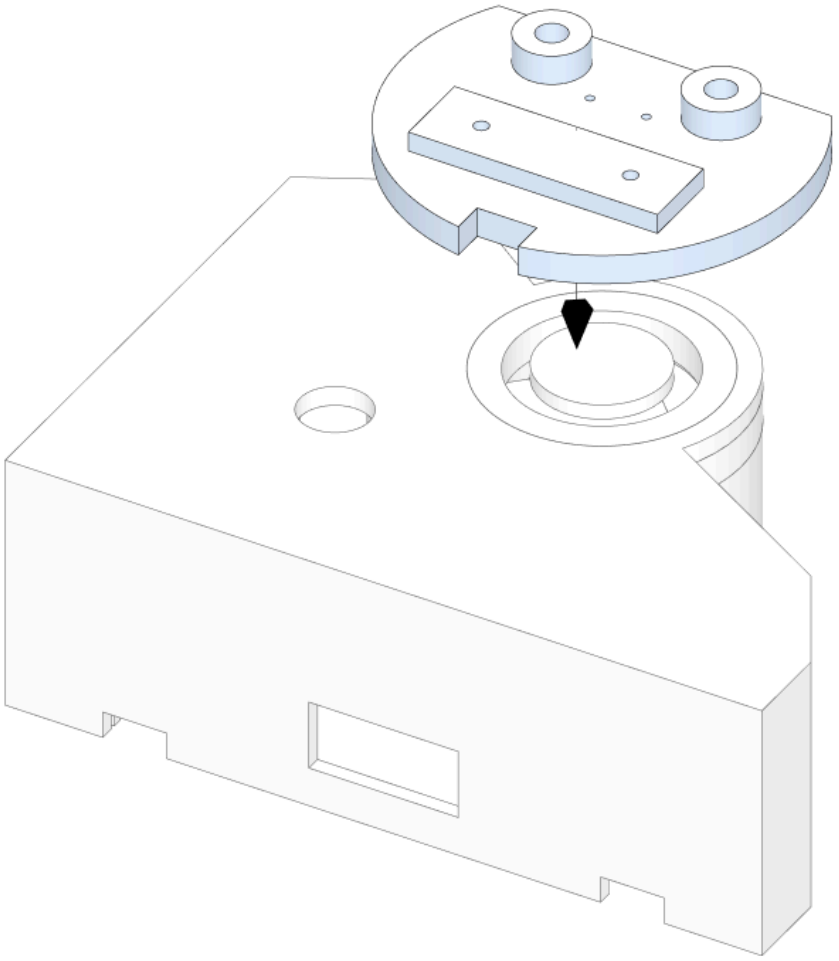
`t 135`

! Set the servo to 135° first (servo_setup: t 135)

5

Turret plate

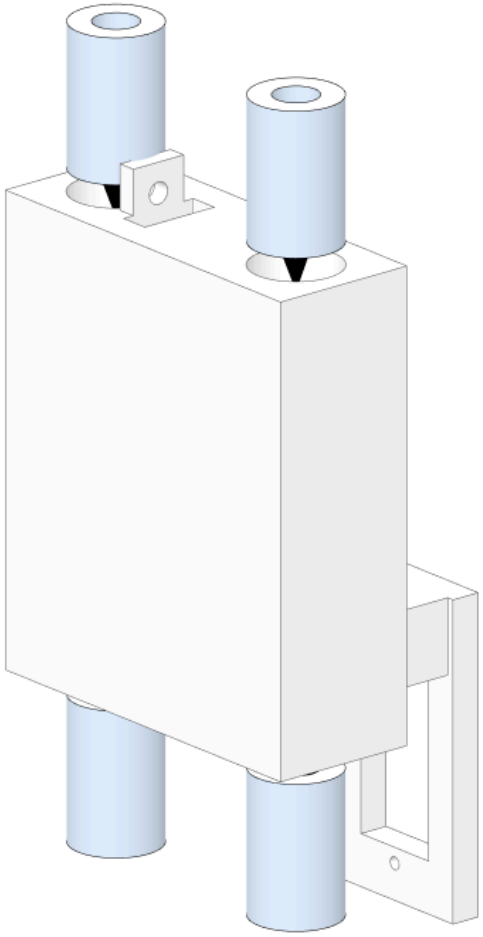
4× servo-bag



Hub into the bearing; rod bosses point at the START can. Turns 90° freely

6

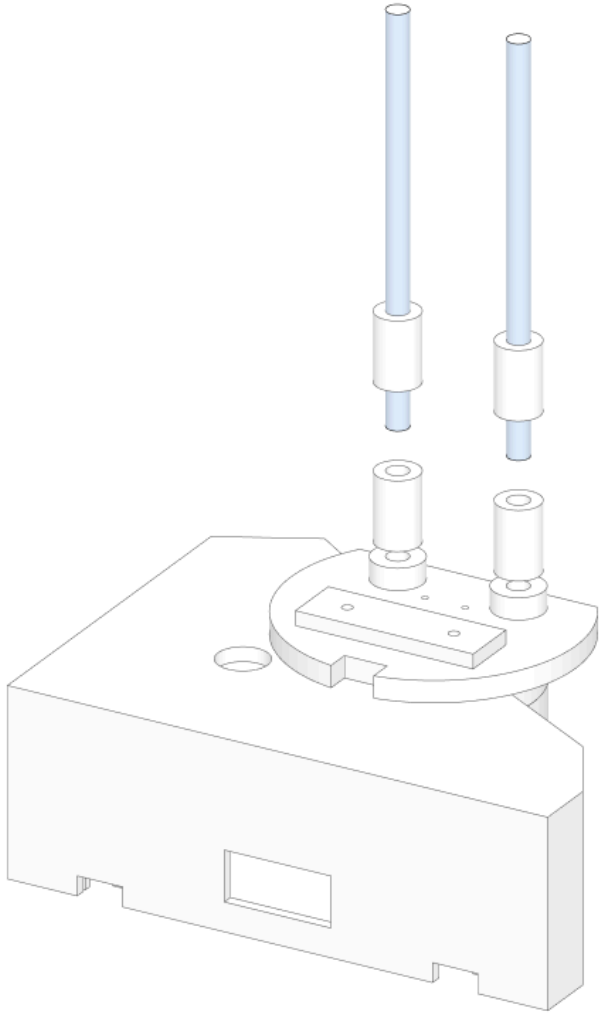
Linear bearings into the carriage



2 from below, 2 from above, flush with the faces

7

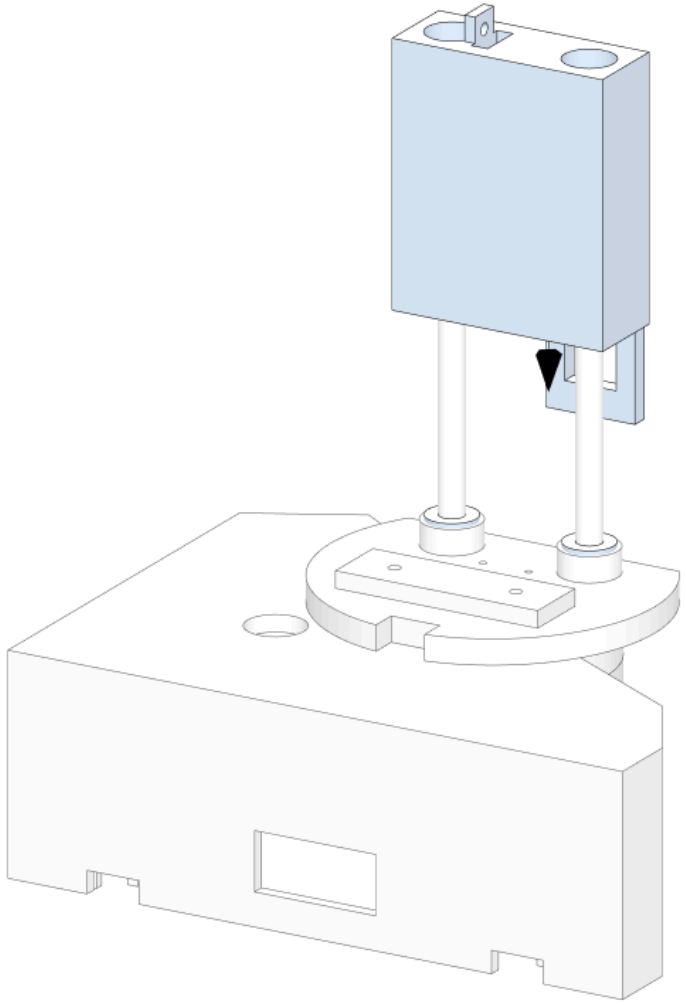
Rods



Press into the blind holes of the turret plate

8

Carriage onto the rods

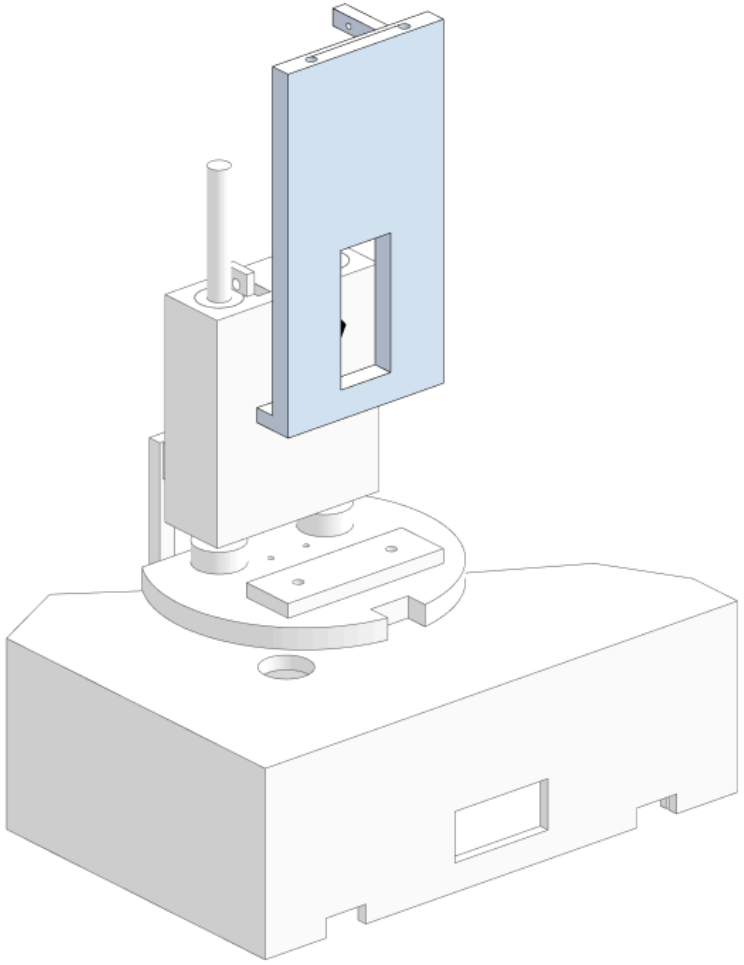


(!) It must slide down under its own weight

9

Spine

2× #6-32 × ½″

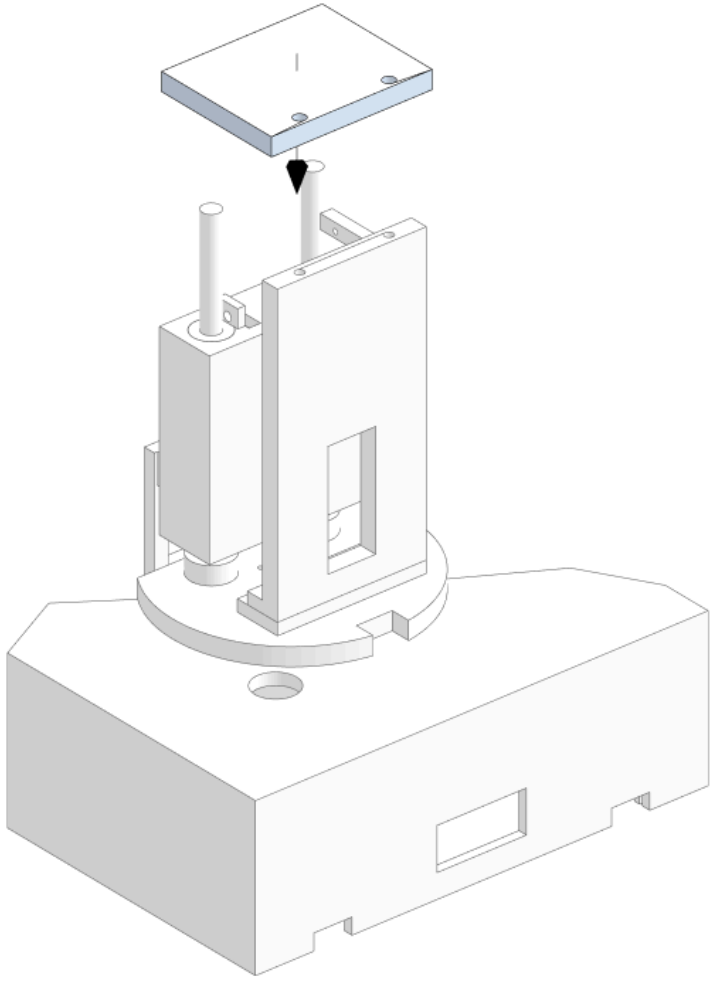


Onto the turret plate behind the rods, screws from above

10

Top cap

2× #6-32 × ½″



Over the rod tops onto the spine. Carriage still slides freely?

11

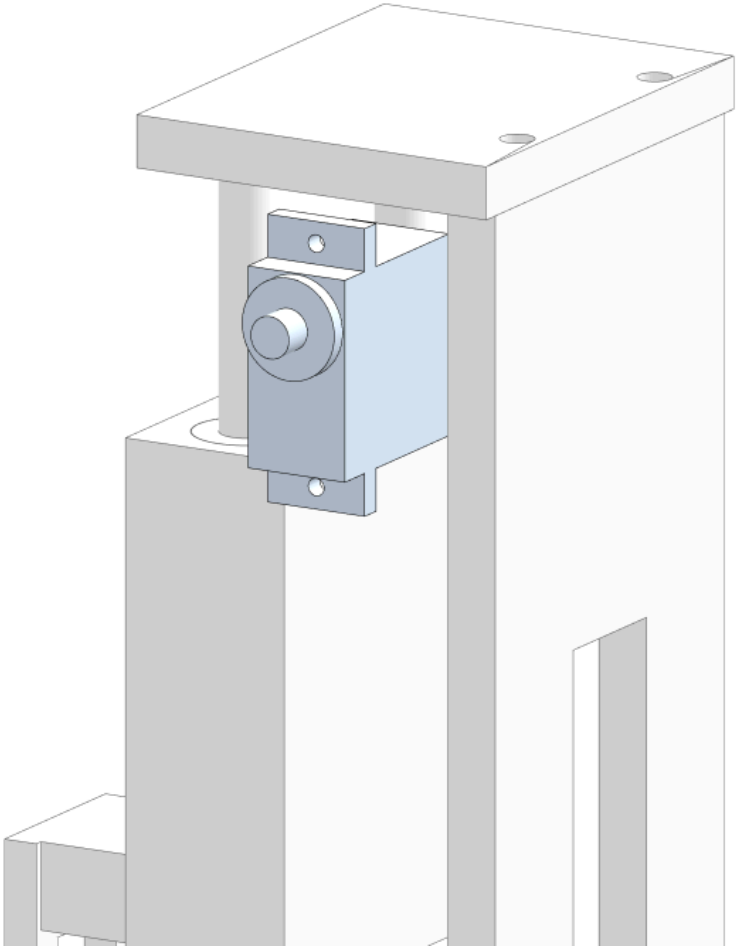
Lift servo

2× servo-bag

MG90S lift

30°

`l 30`

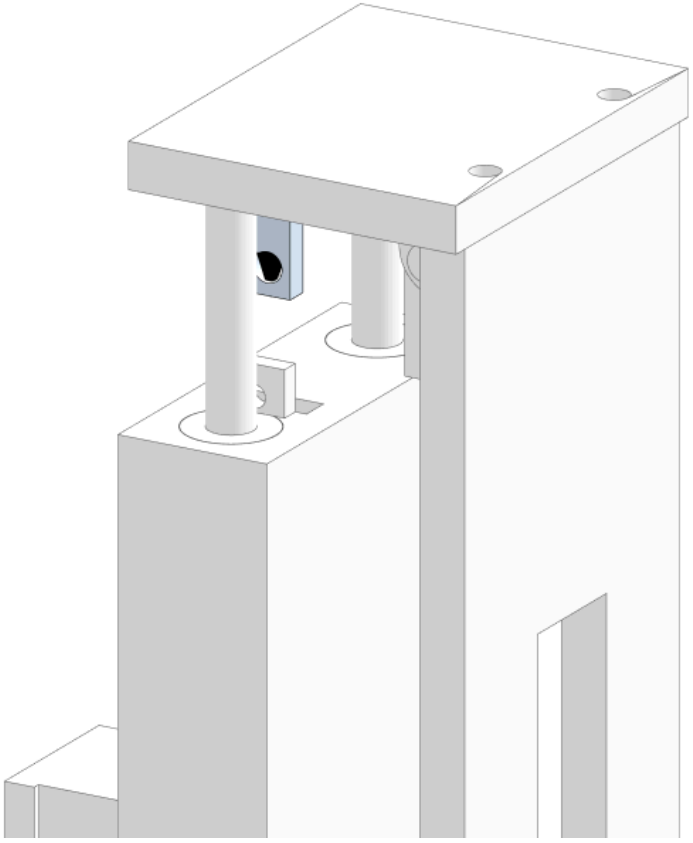


Body through the spine bracket from the left

12

Lift link

1× #6-32 × ½″



! Carriage end: #6-32 as the pivot, don't over-tighten: it must turn

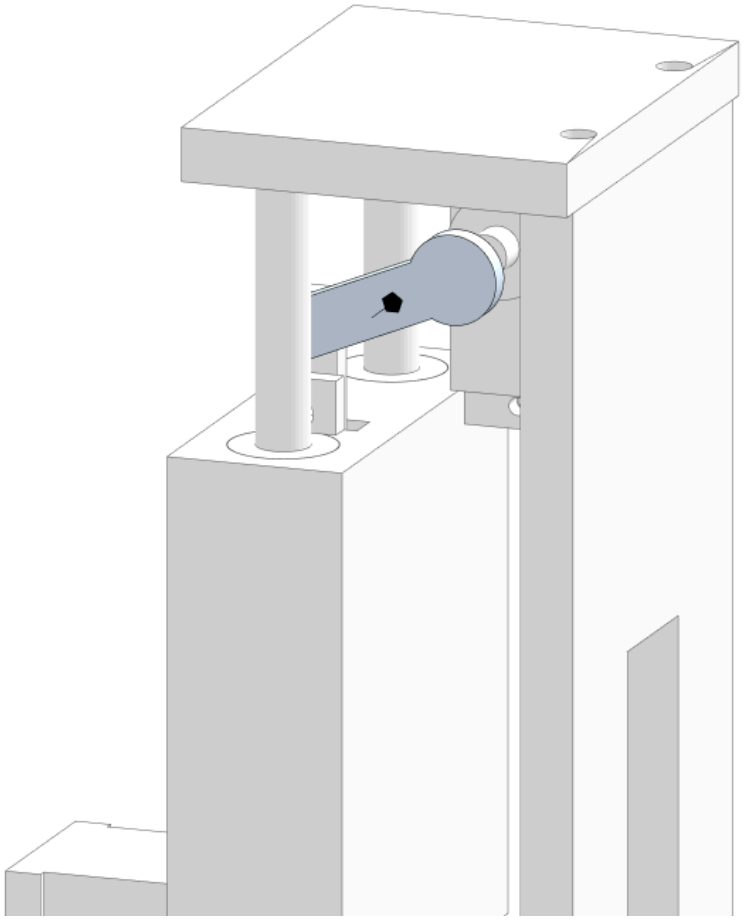
13 Crank

2× servo-bag

MG90S lift

30°

`l 30`



Single-arm horn trimmed to 20 mm, on the spline at 30°; pin screw into the link

14

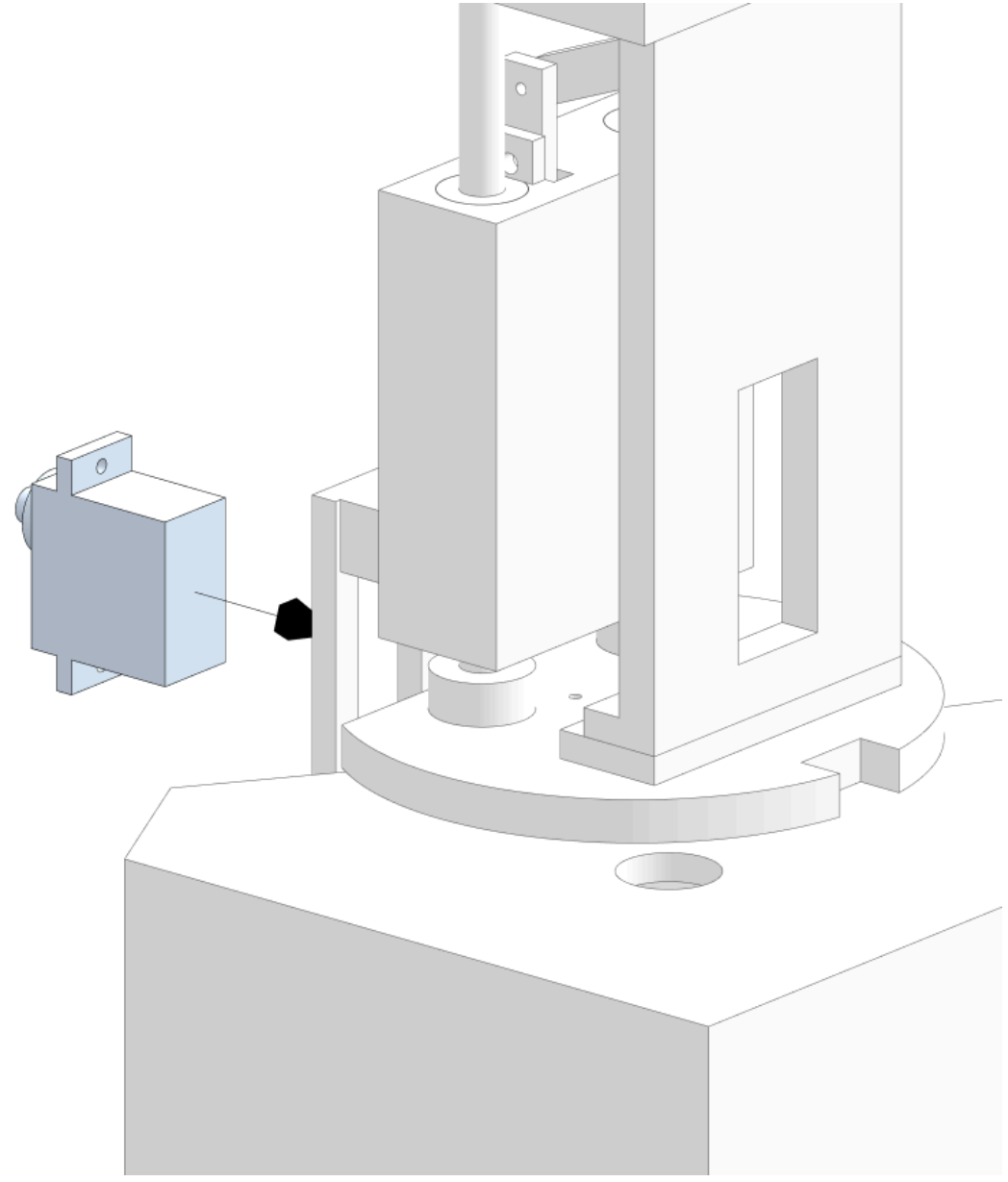
Wrist servo

2×  servo-bag

MG90S wrist

0°

`w 0`



Body through the bracket from the front

15

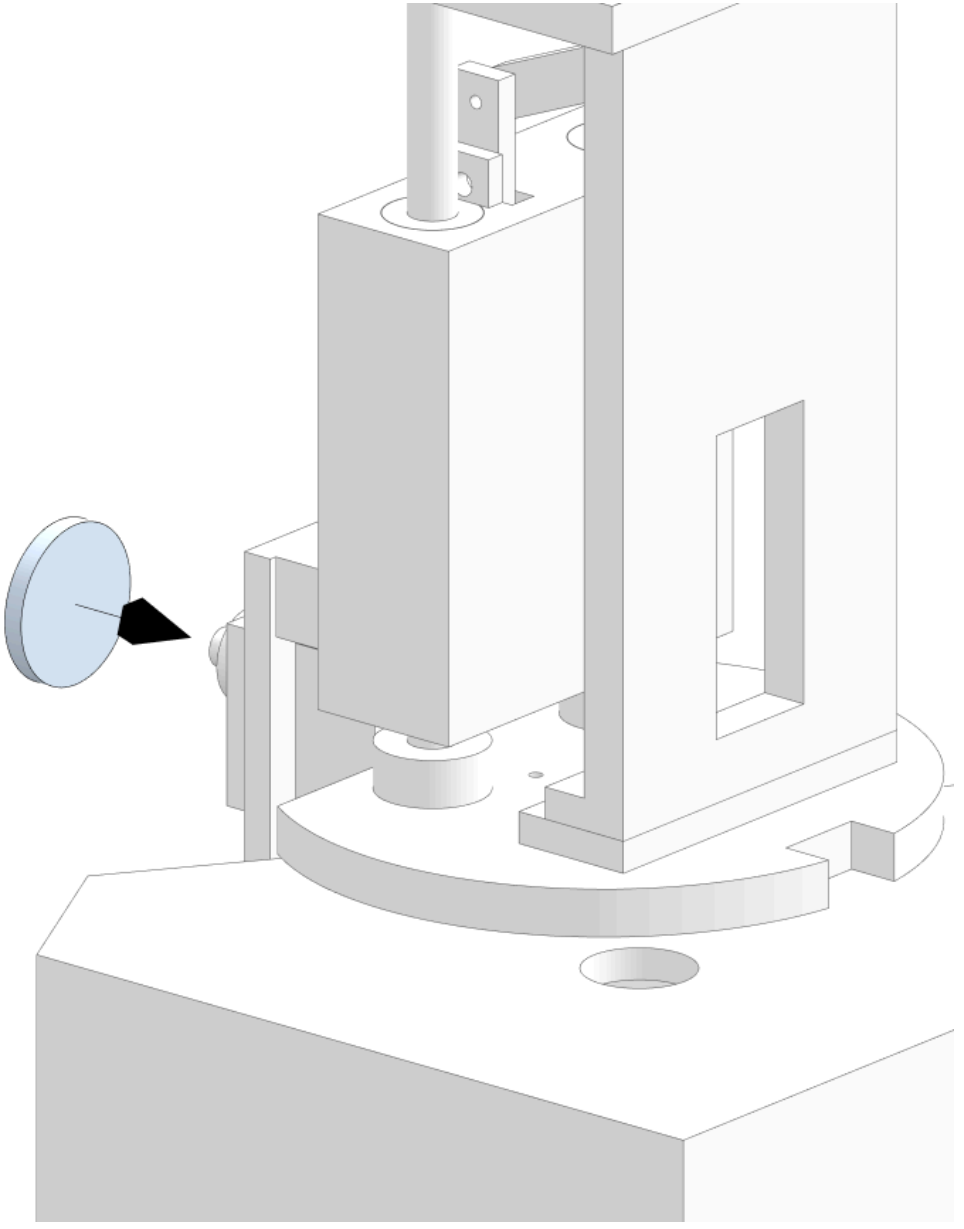
Wrist horn

1×  servo-bag

MG90S wrist

0°

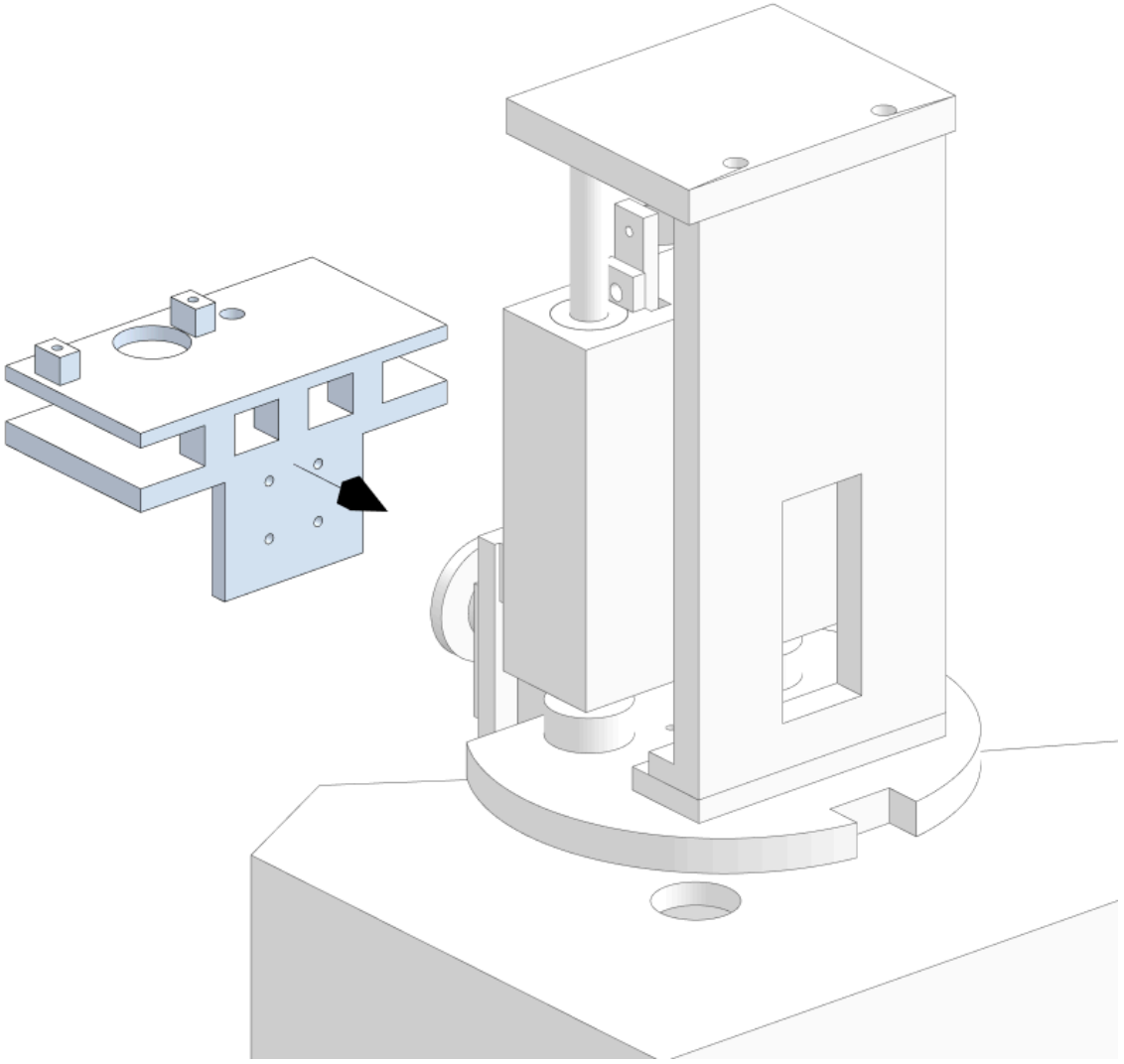
`w 0`



16

Gripper frame

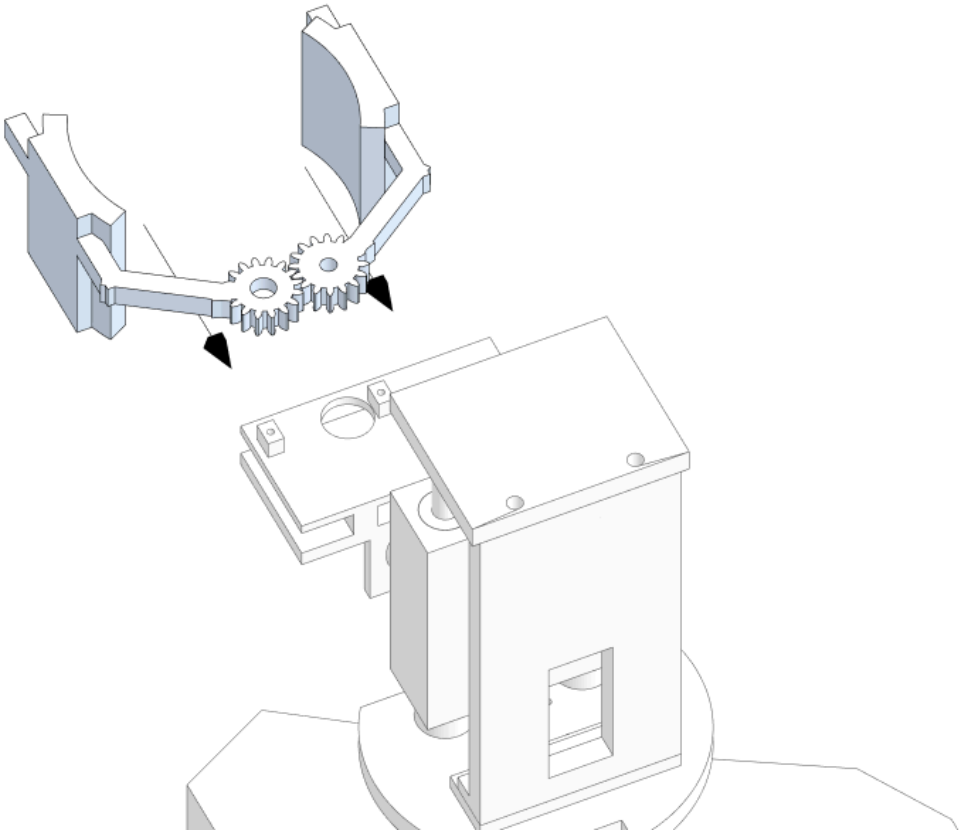
4×  servo-bag



Upright: plates level, opening facing out

17

Both fingers, gears meshed

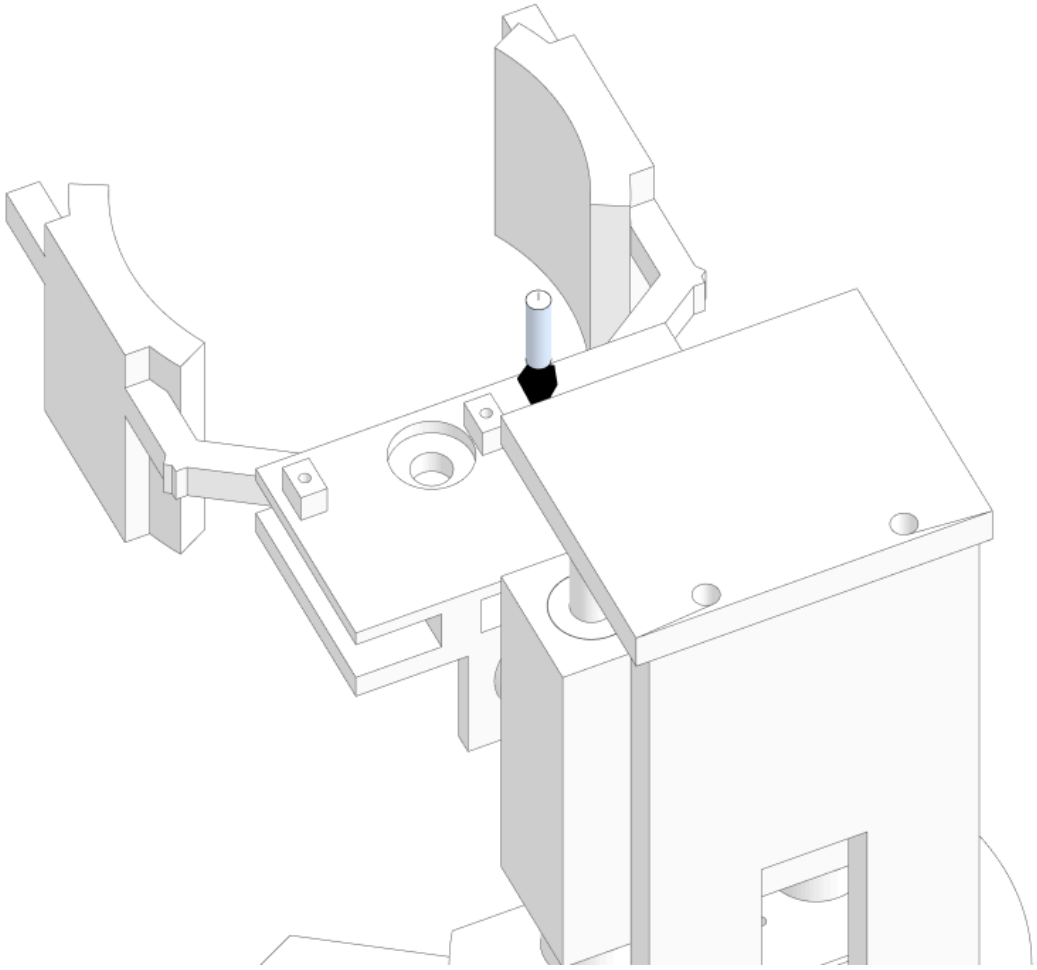


Slide them in together between the plates, open by the same angle

18

Right finger axle

1× #6-32 × ½″



! From the top through the right finger; it must swing freely

19

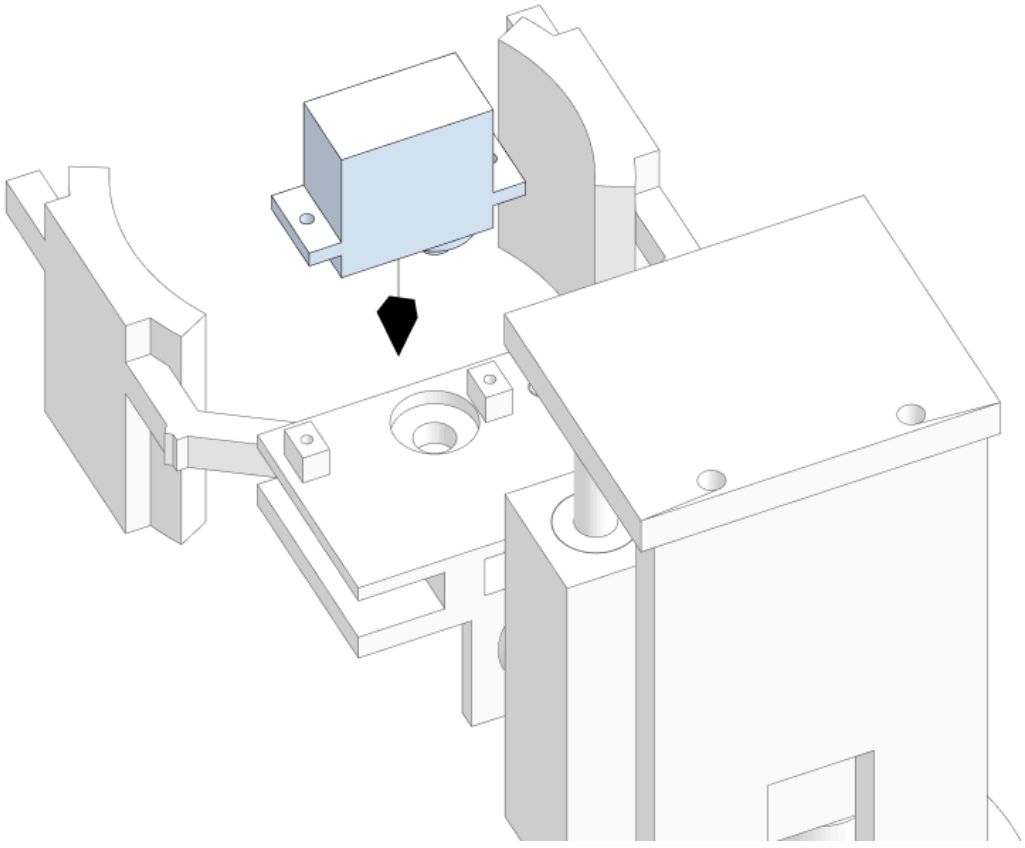
Gripper servo

3× servo-bag

SG90

20°

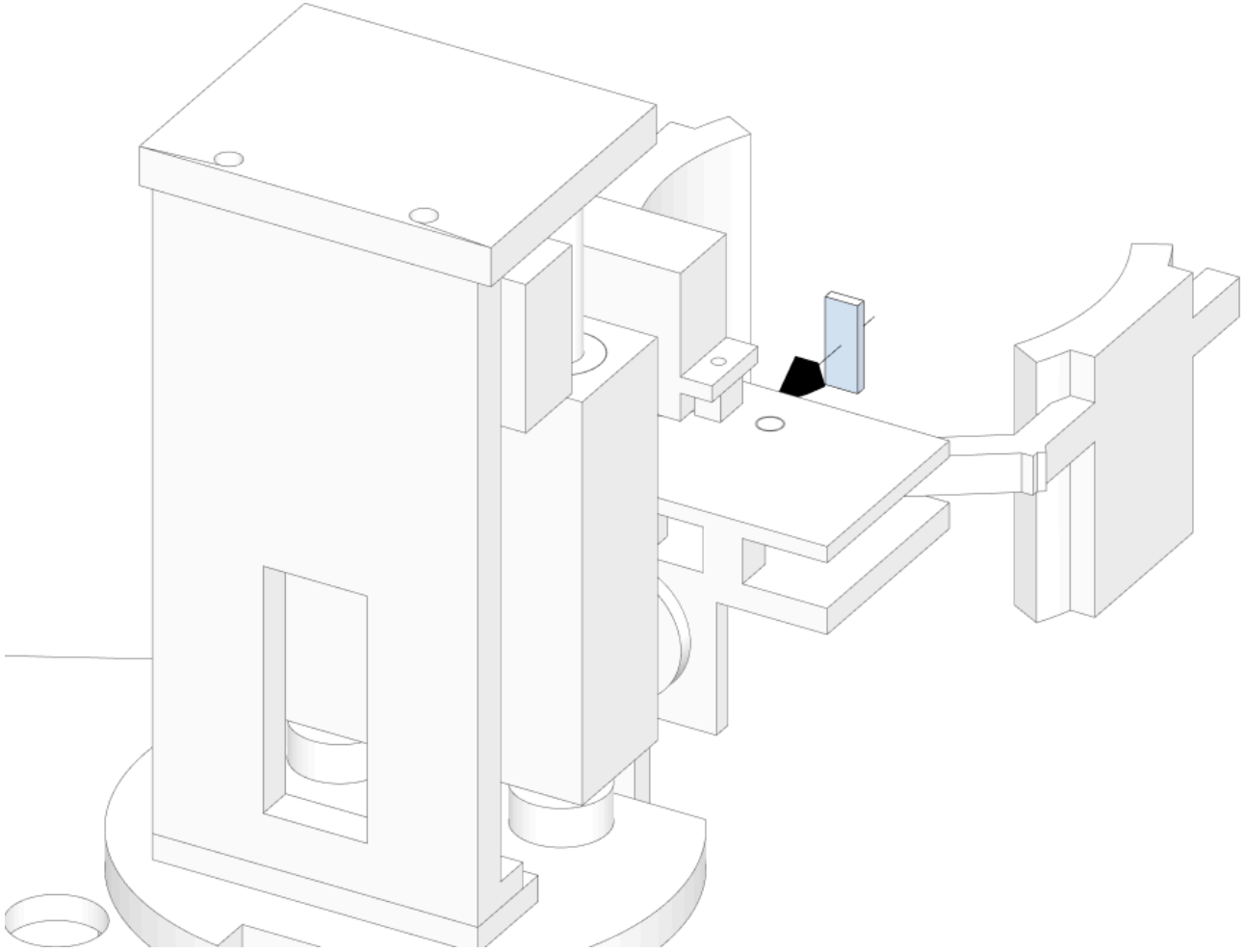
g 20



Spline down into the left finger's socket (it is the horn)

20

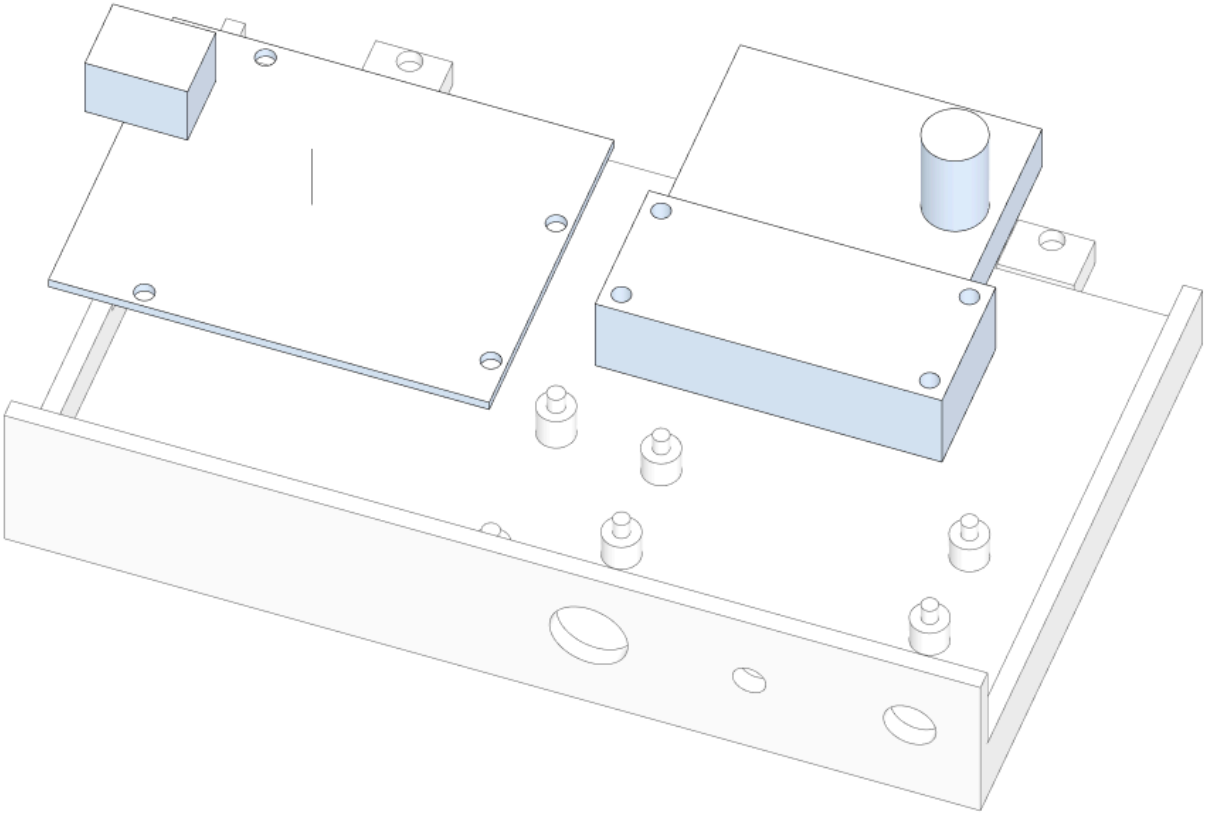
Can bumper



Silicone or rubber, 9 × 13.5 mm, superglue to the frame

21

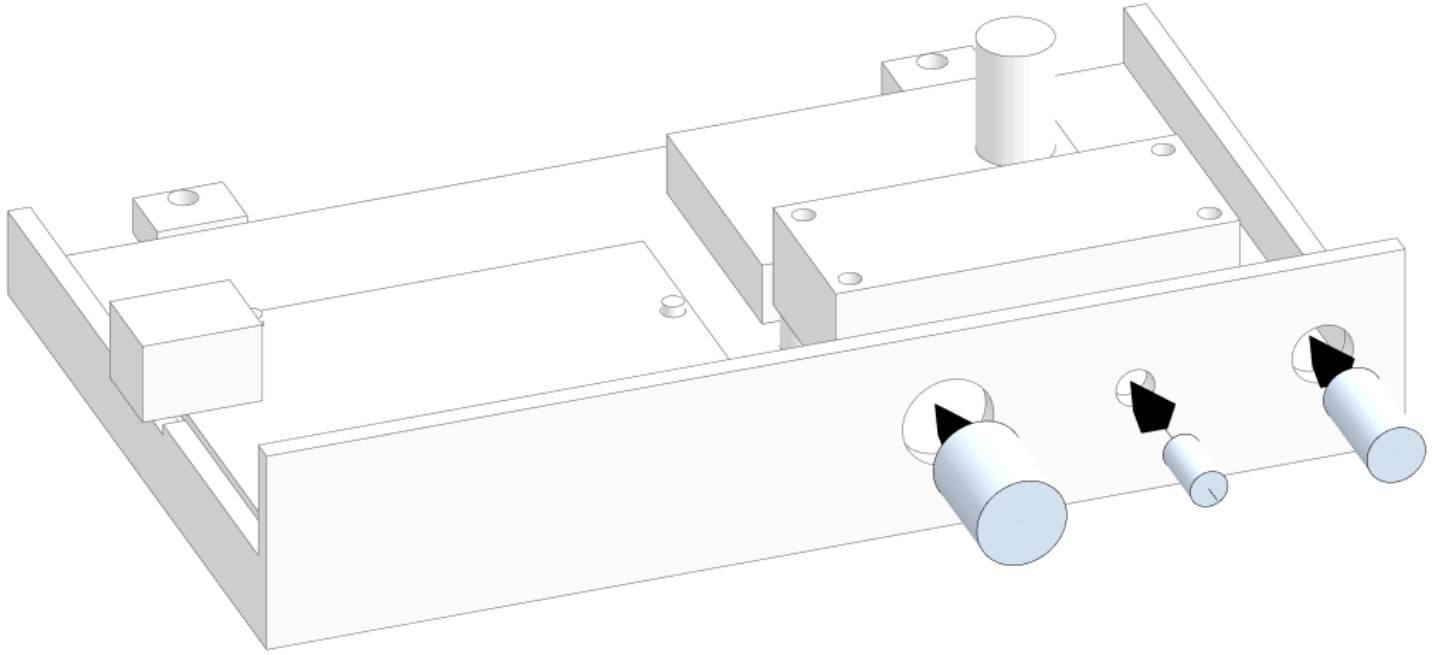
Electronics on the tray



(!) Pegs + a dab of hot glue; breadboard is self-adhesive. Buck already set to 5.8 V

22

Panel: button, LED, power

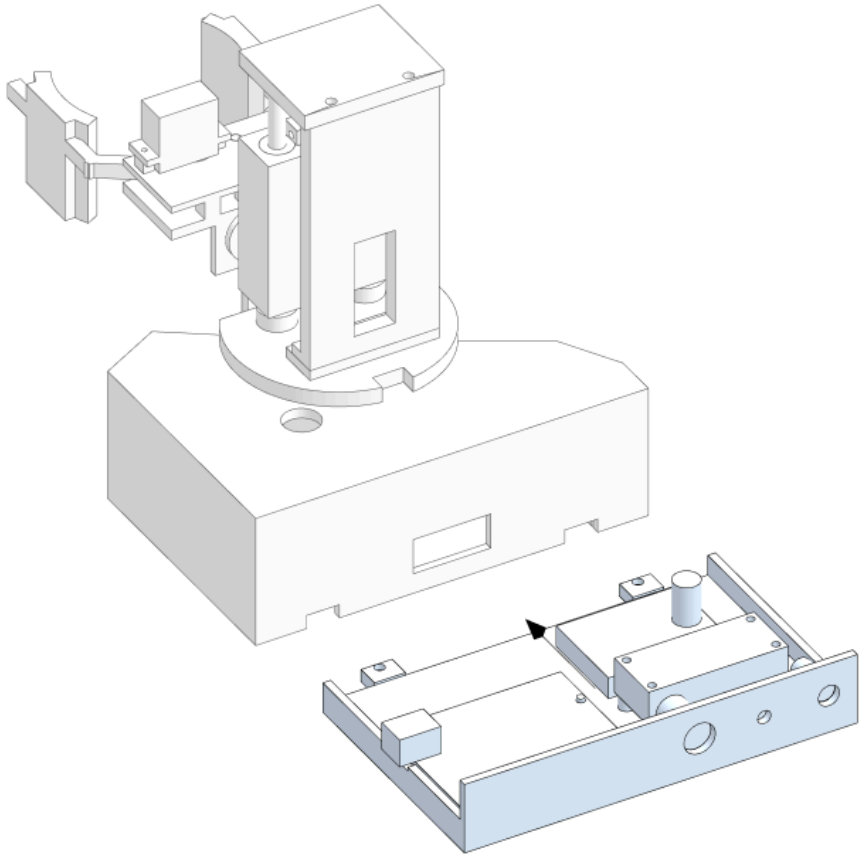


Through the panel holes from the front, nuts inside. Power: DC jack, or the LIZVOR terminal-block tip

23

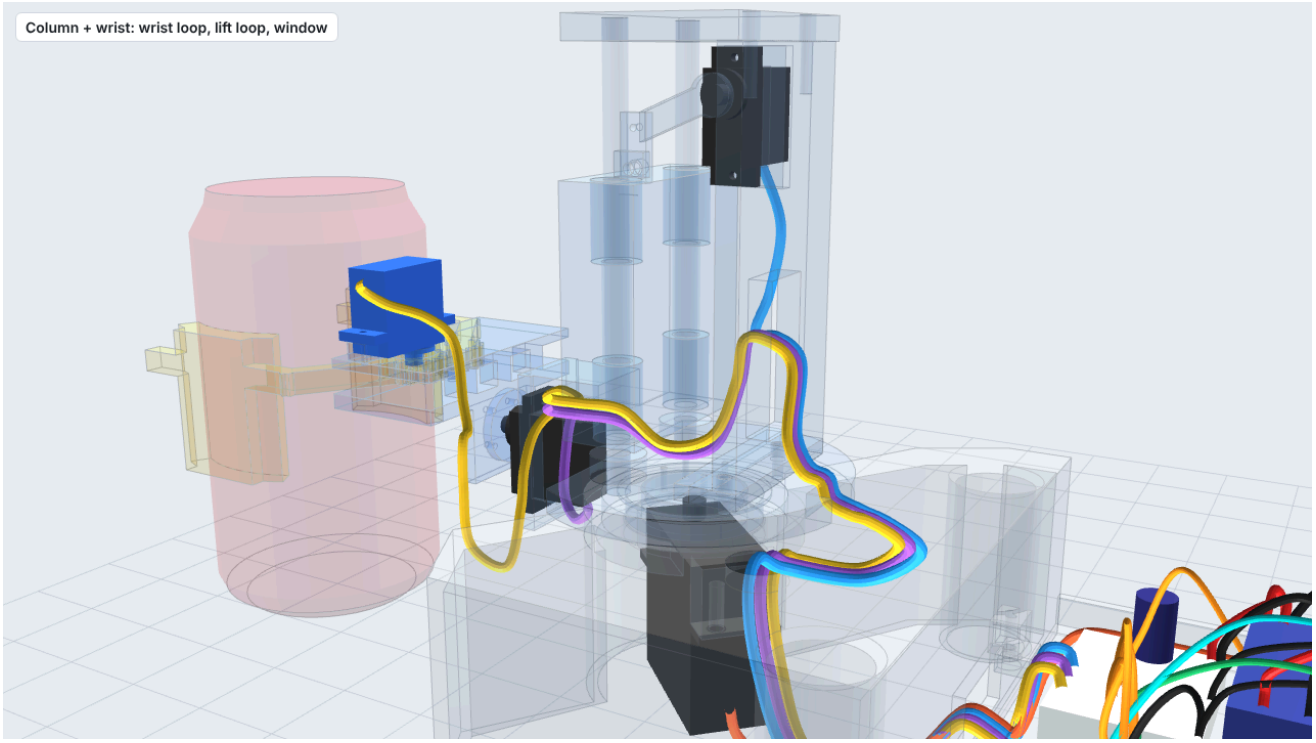
Tray onto the base

2× #6-32 × ½″



Tabs slide under the base wall, screws from below

24 Wiring



Servo lead	Pin
MG996R turntable	D9 (+/− on the buck)
MG90S lift	D10
MG90S wrist	D6
SG90 gripper	D5
START button	D4 + GND
LED + 220 Ω	D13 + GND

Route: claw → wrist loop → along the arm → lift loop → spine window → down the back → turret notch → turn loop → base hole → tray. **Tie at the tie points, leave the 3 loops free.** Wrist loop ≥ 70 mm of cable.

3D view: viewer/wiring_360.html · moves: viewer/wiring_motion.html

(!) Power off while wiring. Servo plugs: brown = GND, red = +5.8 V, orange = signal.

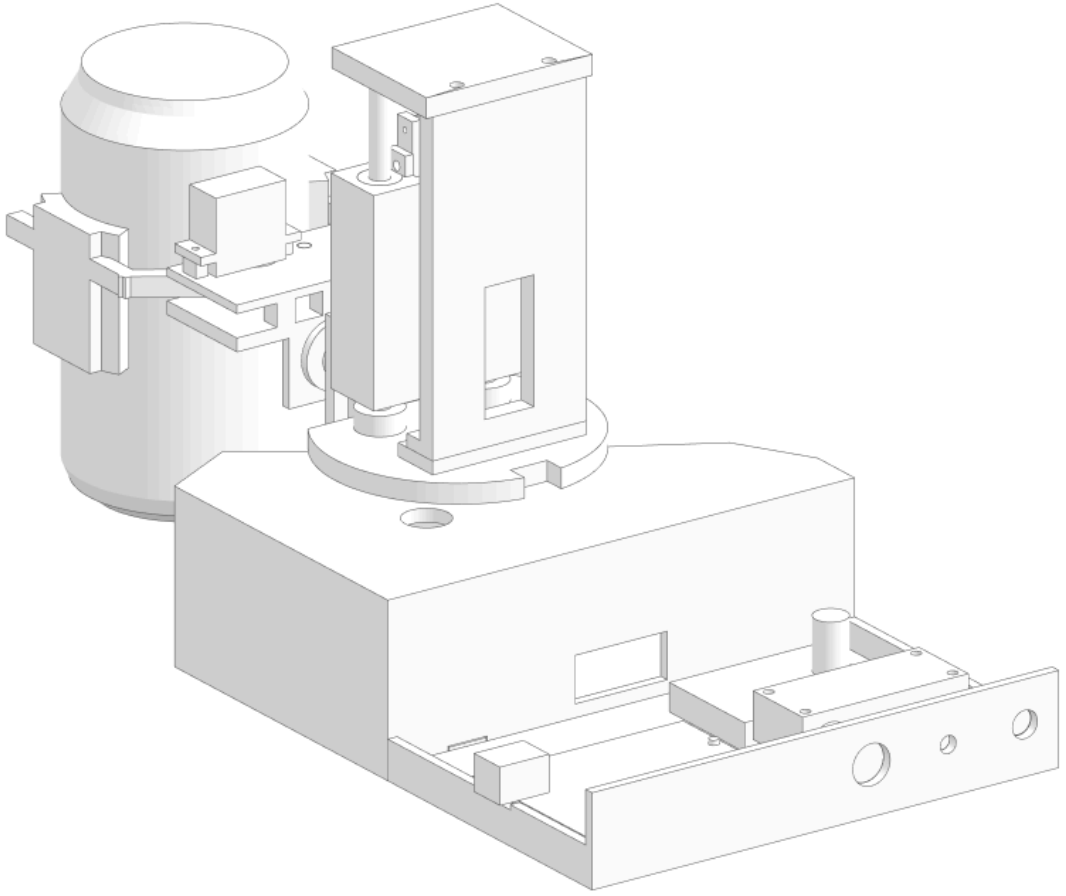
25

Test

1. Upload

`firmware/concept_a`

2. Can on START, 5 in out
3. Press START
4. Can lands upside down at TARGET



If a move is off: re-set its angle with servo_setup, then edit the constant at the top of concept_a.ino.